

Developing Peripherals

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Development Support Engineer

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Product Development

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- Types of Products
- Hardware
- Software
- Q&A

Types of Products

- Modems
  - Analog
  - Digital
  - Wireless
- Scanners, Barcode
  - Add-on
  - Built in

Types of Products (Cont.)

- Keyboards
- Sensors
  - GPS receiver (map position)
  - Magnetic (compass direction)
  - Moisture
  - Pressure
  - Temperature
- Voice recorders

Types of Products (Cont.)

Products best suited for a PC connection

- Printers
  - Print via the desktop
    - Print drivers
- Scanners, document
  - Scan via the desktop
    - Scan drivers
    - Optical Character Recognition (OCR)

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Hardware

- Enabling information
  - Palm Computing's web page (Serial Port Signals)
    - http://www.palm.com/devzone/hw.html
  - Solution Provider Pavilion
    - http://spp.palm.com/enter_password.html
    - Access agreements
      - When approved, access to information
- Serial Port Interface
- IR Port

Enabling Information

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Under Non Disclosure Agreement (NDA)

- Hardware Development Kit (HDK)
  - Serial Port and Modem Casing
    - Obtain proprietary connectors
    - Obtain proprietary cables
    - Have custom cables made
    - Device External Surfaces 3D CAD IGES files
    - Device Drawings
    - Obtain Palm™ Modem casing
    - Obtain Palm V™ Modem casing
    - Have custom case made
    - Modem 3D CAD IGES files

Enabling Information

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- Hardware Development Kit (HDK) (Cont.)
  - Palm IIIx™ Expansion Slot
    - Unique Expansion Slot NDA
      - Memory only
    - Under OEM agreements only
      - Emissive applications

Enabling Information (Cont.)

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Under very selective and strict NDA

- Other 3D CAD IGES files
- Device Charging Specification

Enabling Information (Cont.)

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- Who to contact about
  - Posted Program Agreements
    - Receive on average 117 per day
    - Send it with no modifications
      - Usually processed with 2–3 days of receipt
      - Automated acceptance e-mail sent
    - Send it with modifications
      - Filed in shred bin
      - Automated rejection e-mail sent, no further contact

Enabling Information (Cont.)

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- Who to contact about (Cont.)
  - Special requests to be enabled with information
    - devrelations@palm.com
      - Subject:
        - » HDK
        - » HDK—Charging Spec
        - » CAD
        - » Drawings

Enabling Information (Cont.)

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- Who to contact about (Cont.)
  - Technical question about the information that you have been granted access to
    - devsupp@palm.com
  - Becoming an OEM
    - oeminfo@palm.com
  - Becoming a Licensee
    - licensing@palm.com

Palm OEM/Palm Licensee?

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- OEM
  - OEM R&D time, talent spent on product, solution differentiation that create unique value propositions to target segments
  - Competence is marketing and driving demand
  - Very strong channel relations to the target segment, customers
  - May “own a channel”
  - Strong brand and presence in non-USA countries

Palm OEM/Palm Licensee? (Cont.)

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- Licensee
  - Licensee R&D time, talent spent to “meet or beat” Palm h/w as well as resolve s/w integration
  - Immediate economies of scale with own supply base
  - World class competence in manufacturing of handheld, small form factor devices
  - Global competence in service and support of enterprise or consumer PC/electronics
  - Unique channel with high barriers to entry

Enabling Information (Cont.)

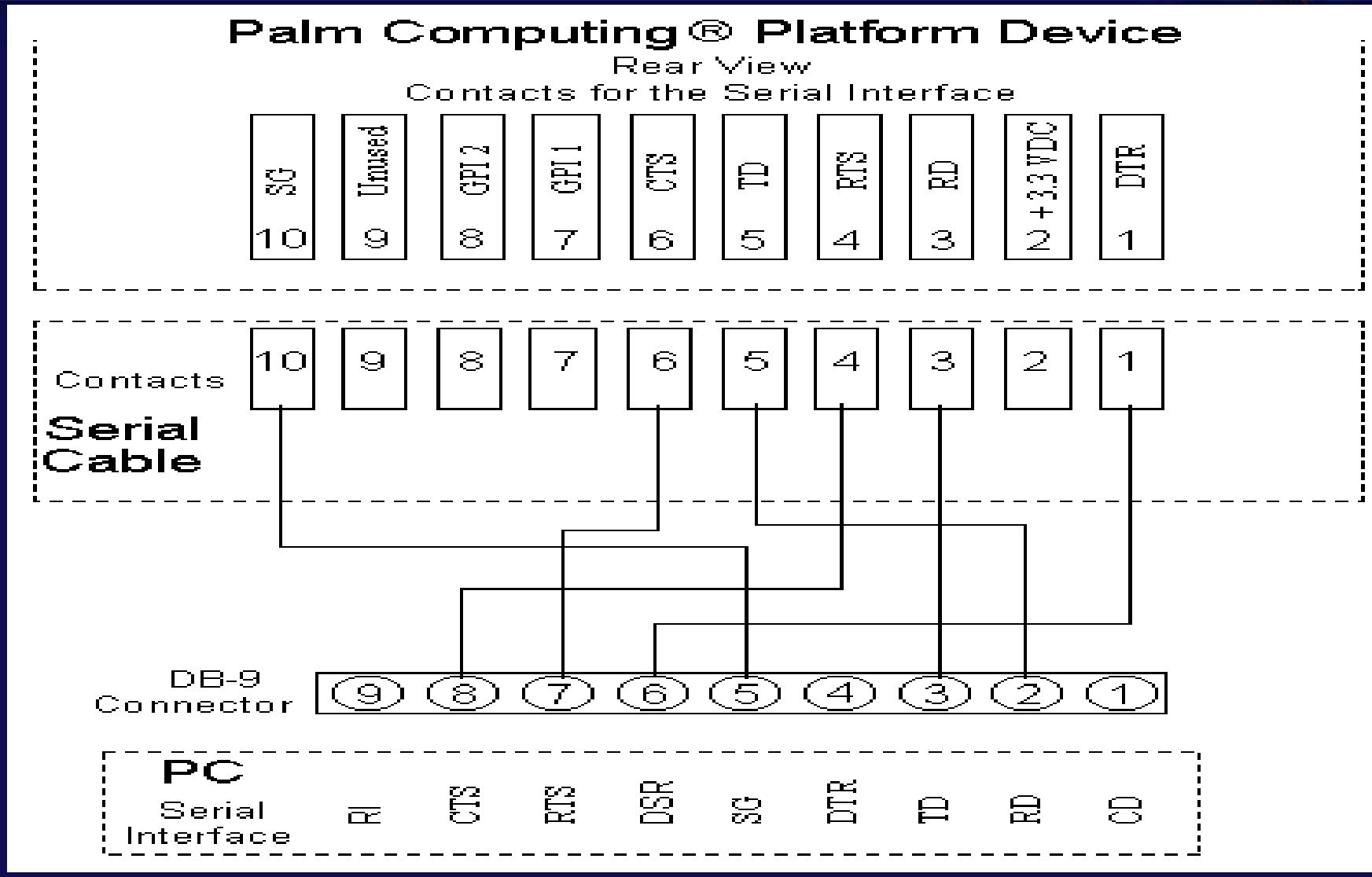
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- Improvements promised and delivered
  - Blanket NDA
    - Serial Port and Modem Casing
    - External surfaces 3D CAD IGES files
  - Create non-confidential information
  - Download information from web page

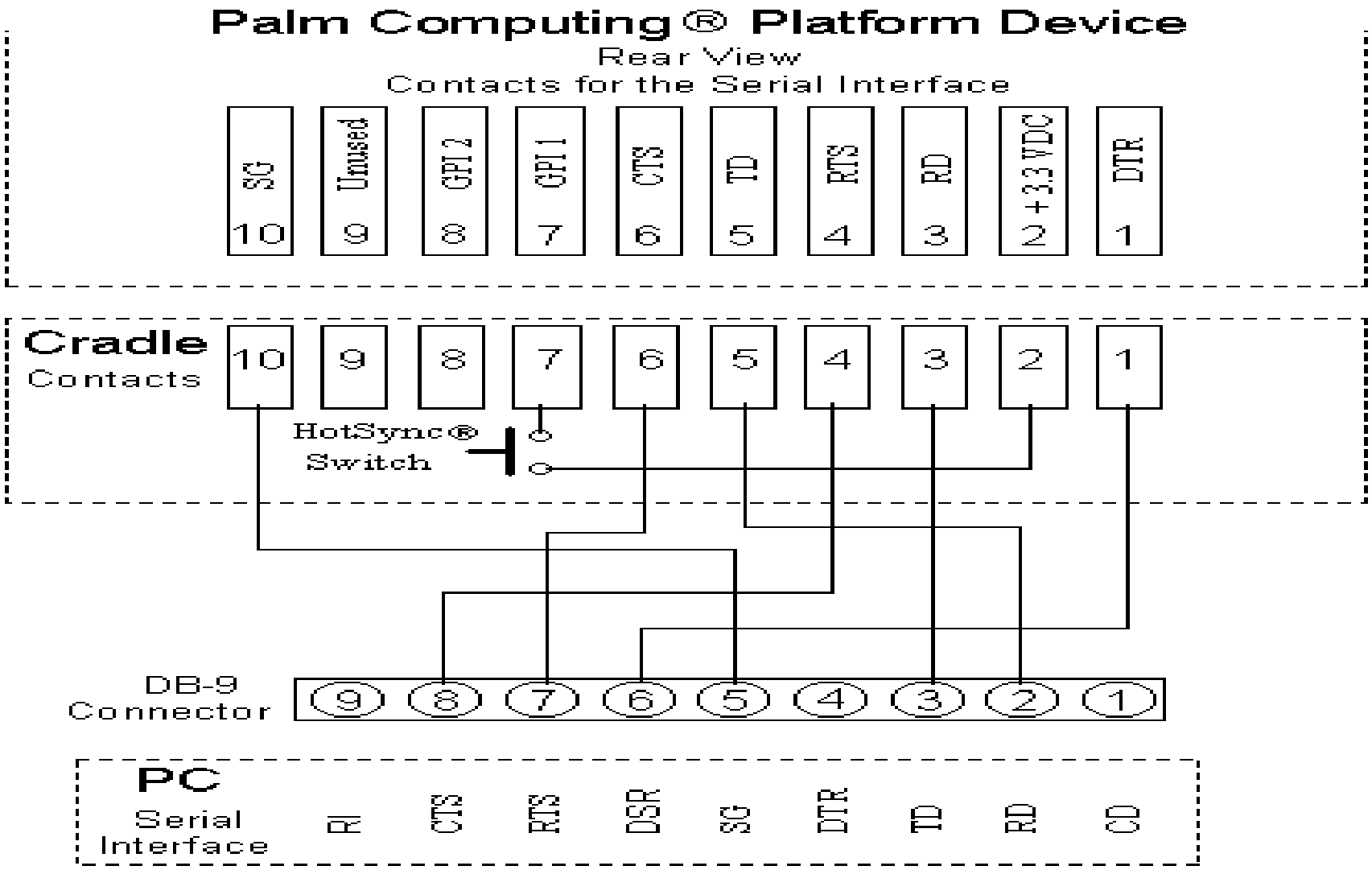
Serial Port Interface

- Palm Computing® platform organizers are not designed to be powered by nor to be the source of power for third party peripherals. Anything that is developed to be connected to the organizer should have its own power source.

Serial Port Interface (Cont.)



Serial Port Interface (Cont.)



Serial Port Interface (Cont.)

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- Palm Computing's proprietary connector
- Use interface latch(es) to attach to organizer
- Electrical signals
  - Non-standard EIA 562
- Palm OS® Serial Manager API

IR Port

- IR devices
  - Palm III, Palm IIIx, Palm IIIe, Palm V, Palm Vx, Palm VII
  - Upgrade card (2 MB memory and IR) for PalmPilot™ device
- Communication protocol
  - IrDA

Product Development

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Software

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Use the API, do not try to talk direct to the hardware

- Present
  - You can only monitor CTS with SerGetStatus
  - No direct signal control
  - Most devices have Flash ROM (upgrade)

Software (Cont.)

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Details of using the API are covered in these sessions

- 186—Debugging Serial Communications
- 128—Adding Communications to Your App/Serial
- 122—Beaming Data with the Exchange Manager
- 280—Debugging Serial Drivers

Summary

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- Enable peripheral developers
- Free access to custom tooling (NDA)
- Stay within Palm OS® API

What You Can Do Today

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- As developers did over the past year …go forth and develop useful hardware products to attach to Palm Computing® platform organizers!

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